BIZORA AI

● PRODUCTION LIVE ON RENDER

Omni-Mind Autonomous Retail Intelligence & Multi-Store ERP Operating System

An enterprise-grade, multi-tenant retail operating system designed for modern multi-store owners, department managers, cashiers, suppliers, and customers. Bizora replaces fragmented legacy retail tools with an intelligent, autonomous core orchestrating real-time POS billing, inventory 360, AI decision automation, double-entry financial ledgers, and multi-portal supply chain synchronization.

1. LIVE ACCESS DIRECTORY & DEPLOYED PRODUCTION LINKS

All five client portals and the centralized NestJS backend are deployed live on Render with automatic HTTPS/TLS, automated environment orchestration, and real-time health telemetry. Click any link below to access the live environments:

PORTAL / SERVICE	LIVE INTERACTIVE URL	PRIMARY TARGET	CORE CAPABILITY
👑 **Store Owner Portal**	`bizora-owner.onrender.com`	BUSINESS OWNERS	Executive P&L, POS, 360 Inventory, AI Decisions
🛡️ **Super Admin Portal**	`bizora-admin.onrender.com`	PLATFORM ADMINS	Tenant Approvals, Health Monitor, Audit Logs
🚚 **Supplier Hub**	`bizora-supplier.onrender.com`	VENDORS & SUPPLIERS	Inbound PO Orders, Fulfillment & Dispatches
🛍️ **Customer CRM Portal**	`bizora-customer.onrender.com`	END CONSUMERS	Digital Receipts, Loyalty Points & Order History
🌐 **Web Marketing & Tour**	`bizora-web.onrender.com`	PUBLIC VISITORS	Platform Overview, Pricing & Self-Onboarding
⚙️ **Centralized API Microservice**	`bizora-backend-gtoe.onrender.com`	BACKEND SERVICE	NestJS REST Engine, Prisma, PostgreSQL Pool
💓 **API Health & Monitoring**	`.../api/health`	DEVOPS / PING	5-Minute Automated Heartbeat & Resend Alerts
🐙 **Source Code Repository**	`github.com/omkhade75/project-h`	GITHUB MONOREPO	Zero-rebase Lovable & Render Synced Branch

2. THE CORE BUSINESS PROBLEM & THE BIZORA SOLUTION

❌ The Traditional Retail Nightmare

- **Disconnected Silos:** POS billing is disconnected from accounting ledgers, requiring manual end-of-day Excel reconciliations.
- **Inventory Shrinkage & Waste:** Perishable products expire unnoticed on shelves; stock-outs occur during peak shopping hours.
- **Data Leakage & Multi-Store Confusion:** Multi-branch owners mix up store inventories, receipts, and cash flow across locations.
- **Passive Customer Retention:** Zero tracking of customer lifetime value (LTV), churn risk, or automated post-purchase re-engagement.
- **Manual Vendor Restocking:** Phone calls and lost paper purchase orders cause shipment delays and inaccurate deliveries.

✅ The Autonomous Bizora Operating System

- **Strict Multi-Tenant Isolation:** Store 1, Store 2, and Store 3 operate with mathematical data isolation and dedicated dashboards.
- **Autonomous AI Replenishment:** Dynamic reorder points trigger automated Purchase Orders to suppliers before stock reaches zero.
- **Real-Time Financial Integrity:** Every sale automatically updates the General Ledger, Tax (GST/VAT) provision, and net profit in real time.
- **AI Voice & WhatsApp Omnichannel:** Integrated Vapi AI voice agents and WhatsApp receipting keep customers and vendors connected.
- **Self-Healing Health Monitoring:** Real-time background error reporting automatically alerts the engineering team with zero silent crashes.

3. MULTI-STORE TENANT ISOLATION & SECURITY ENGINE

A fundamental architectural requirement of Bizora AI is **Strict Multi-Tenant Isolation**. Store 1, Store 2, and Store 3 must never have cross-contaminated data. An unapproved or random email can never access a store's financial ledgers or customer records.

1. Registration & Approval

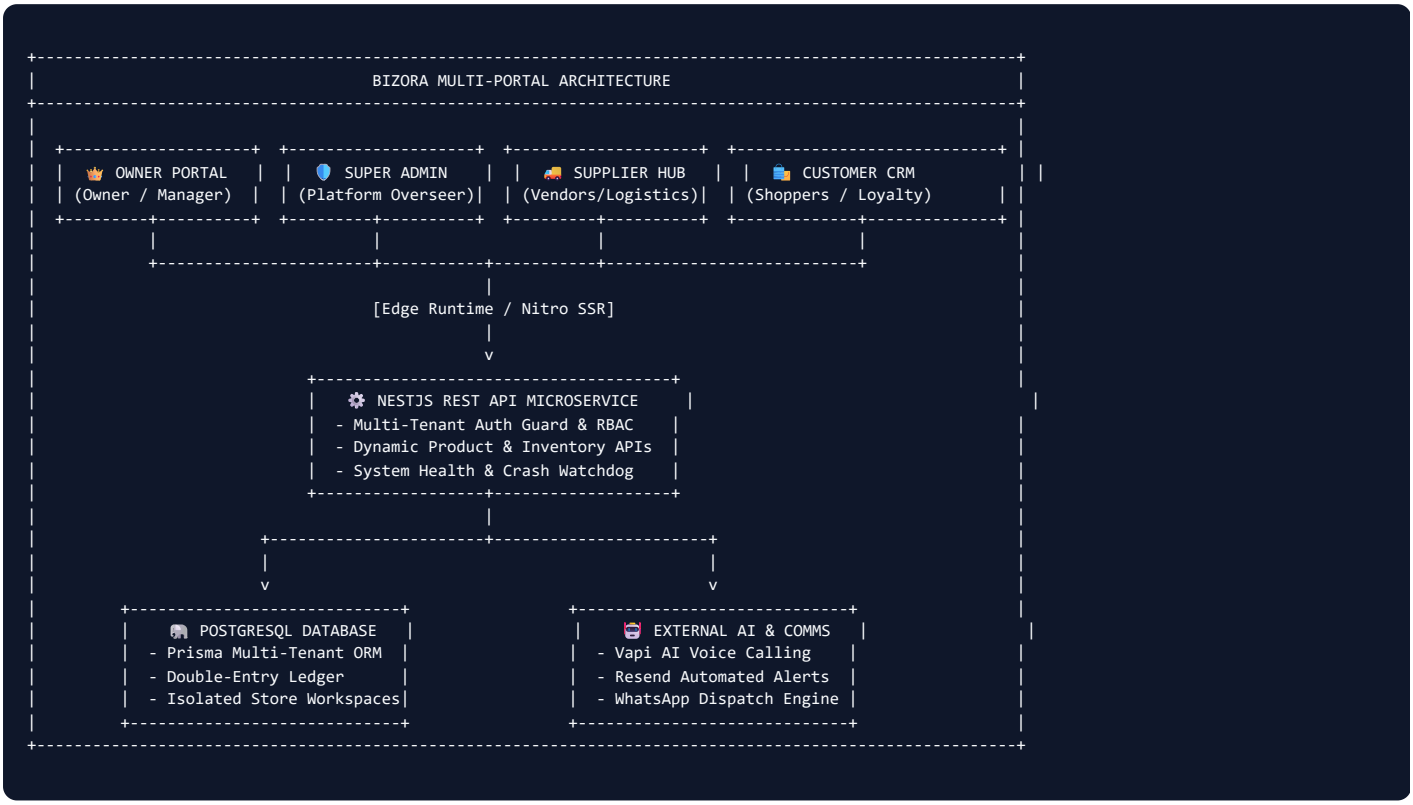
New store owners submit company credentials via the registration portal. The account is placed in a sandboxed, pending state until verified by the Super Admin.

2. Tiered Dashboard Provisioning

Upon admin approval, the system provisions an isolated workspace initialized to the subscribed plan (Starter, Pro, Enterprise) with an empty, clean dataset.

3. Tenant ID Scope Binding

Every database transaction, query, and invoice is strictly scoped to the store's cryptographic workspace ID. Cross-tenant reads are prevented at the ORM layer.



4. CORE APPLICATION MODULES & USER JOURNEYS

📊 Executive Command Center & Live Ops

Real-time ticker tracking gross revenue, footfall velocity, live transactions, and floor activity radar. Includes the emerald real-time sync beacon signaling zero-refresh background updates.

📦 Smart Inventory 360 & Expiry Watchdog

Full SKU lifecycle management with barcode lookup, batch tracking, unit economics, reorder threshold alerts, and early-warning expiry dashboards to eliminate perishable write-offs.

⚡ High-Velocity POS & WhatsApp Invoicing

Lightning-fast point-of-sale interface supporting barcode scanning, discounts, multi-tender payments (UPI, Cash, Card), and one-click WhatsApp digital receipt dispatch.

👥 Customer 360, Loyalty & Voice AI

Tracks customer lifetime value (LTV), visit frequency, loyalty tiers (Gold, Silver, Platinum), and triggers outbound conversational Vapi AI voice agents for abandoned carts and VIP offers.

🚚 Supplier Hub & Automated Purchase Orders

Direct supplier collaboration portal. Generates automated purchase orders when stock depletes, monitors vendor lead time, and tracks on-time delivery percentages.

💰 Financial Intelligence, P&L & GST/Tax

Automated double-entry General Ledger accounting. Real-time income statements, categorized expense management, capital investment tracking, and automated GST/VAT tax reporting.

5. COMPLETE TECHNOLOGY STACK & IN-DEPTH TECHNICAL RATIONALE

Every library, framework, and cloud service in Bizora AI was deliberately chosen to satisfy strict enterprise requirements: zero hydration lag, compile-time type safety, resilient database isolation, and instant real-time synchronization.

1. React 19 + TanStack Start & TanStack Router

FRONTEND & ROUTING FRAMEWORK

Why Chosen: TanStack Router provides 100% end-to-end type-safe routing, type-checked search params, and parallel route data loaders. In a retail application with deep nested views (e.g. `/products?filter=low-stock&store=2`), type-safe search params eliminate URL parsing bugs. TanStack Start combines client-side interactivity with lightweight Server-Side Rendering (SSR) for instant first-contentful paint without hydration mismatches.

2. Nitro Serverless Engine & Vite 8

BUILD TOOL & EDGE SERVER RUNTIME

Why Chosen: Vite 8 delivers sub-second Hot Module Replacement (HMR) and ultra-optimized production rollups. Nitro packages each portal into a portable, cloud-agnostic server worker capable of deploying directly to Render, Cloudflare Workers, or standard Node environments with zero code changes and minimal cold-start latencies.

3. Tailwind CSS v4 & Radix UI Primitives

DESIGN SYSTEM & COMPONENT ARCHITECTURE

Why Chosen: Tailwind CSS v4 provides a zero-runtime CSS engine with modern OKLCH color space support for vibrant, high-contrast dark and light themes. Radix UI supplies unstyled, accessible (WAI-ARIA compliant) headless primitives (Dialogs, Slide Drawers, Popovers, Tabs, Tooltips) that guarantee keyboard navigation and screen-reader accessibility while allowing custom smooth CSS animations.

4. NestJS (Node.js & TypeScript)

ENTERPRISE BACKEND MICROSERVICE

Why Chosen: NestJS enforces an enterprise architectural pattern using Dependency Injection (DI), modular providers, and strictly typed request lifecycle pipelines (Guards, Interceptors, Pipes). It guarantees that authentication, multi-tenant workspace scoping, and business logic remain cleanly decoupled and easily testable.

5. PostgreSQL Database & Prisma ORM

DATABASE & OBJECT-RELATIONAL MAPPING

Why Chosen: Financial ledgers, stock counts, and sales receipts require strict ACID transaction guarantees that NoSQL databases cannot reliably provide. PostgreSQL provides rock-solid relational integrity and indexing. Prisma ORM provides compile-time query generation, automated database migrations, connection pooling, and complete protection against SQL injection attacks.

6. TanStack React Query

SERVER STATE SYNCHRONIZATION & CACHE

Why Chosen: Manages complex server caching with automatic stale-while-revalidate background synchronization, optimistic mutations (instant UI updates when an item is scanned or sold), and automatic retries on network blips. This allows the dashboard to reflect real-time store changes without manual page refreshes.

7. Vapi AI & Resend Email Infrastructure

VOICE AI & AUTOMATED TELEMETRY ALERTS

Why Chosen: Vapi AI integrates ultra-low latency conversational AI voice agents directly into the CRM and Supplier portals for automated phone follow-ups, reorder confirmation, and customer feedback. Resend delivers high-deliverability transactional emails for instant administrative alerts, company approval notifications, and automated 24/7 crash reporting.

8. Recharts & D3 Data Visualization

FINANCIAL ANALYTICS & CHARTING

Why Chosen: Declarative SVG-based charting that renders smoothly on both desktop and mobile devices. Enables interactive sparklines, multi-period revenue area charts, expense category breakdowns, and predictive demand forecast trajectories.

9. Render Cloud Platform

PRODUCTION HOSTING & CI/CD PIPELINE

Why Chosen: Seamless Git-backed continuous deployment pipeline that automatically detects branch pushes, manages environment secrets, provisions zero-downtime rolling deployments, and provides SSL termination and HTTP/2 out of the box.